STHORAM BMS

Technical White Paper — Volume II

Cell Isolation in $LiFePO_4$ Battery Strings:

The Electrochemical Consequences of the Connected Idle State and the Engineering Case for Complete Inter-Cell Electrical Isolation at Rest

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Abstract

Peer-reviewed research has established that battery degradation does not pause when a battery is idle. The foundational calendar-aging study in the Journal of The Electrochemical Society (Keil et al., 2016) showed that capacity fade accumulates continuously during storage under open-circuit conditions, driven by parasitic reactions at the electrode–electrolyte interface — chiefly SEI growth on the graphite anode — with the fade governed primarily by loss of cyclable lithium inventory **[Ref 01]**. Because idle intervals are, in most applications, far longer than operating intervals, calendar aging has a substantial impact on total battery life. Experimental work published in Applied Sciences (MDPI, 2025) confirmed for LFP batteries specifically that calendar aging is strongly governed by state of charge and temperature, with higher SOC and elevated temperature accelerating capacity fade via enhanced SEI growth **[Ref 02]**.

This paper is precise about what those findings do and do not imply. Calendar aging is an intra-cell electrochemical process: it proceeds inside each cell whether the string is connected or not, and no switching architecture can stop it. What the connected idle state adds — and what isolation removes — is a set of distinct, system-level hazards: (i) external parasitic drain paths through monitoring and balancing electronics, which discharge cells slowly and, critically, unevenly, so that cells drift apart in SOC across long rest periods and the balanced state established before rest is progressively lost **[Ref 03 — simulation study; Ref 12]**; (ii) the risk that any sustained current path — a connected quiescent load, a monitoring fault, or leakage — drives the weakest cell toward deep over-discharge, which peer-reviewed research in Scientific Reports (2016) showed causes irreversible copper current-collector dissolution and internal short-circuit formation beyond approximately −12% SOC **[Ref 05]**; (iii) the continuous presence of the full stacked string voltage — approximately 77 V for a 24S LFP string — across the internal series chain, sustaining insulation stress, arc-fault energy, and external-short-circuit fault energy at all times **[Ref 13]**; and (iv) the electrical dimension of fault propagation, documented by Sandia National Laboratories, whose module-level experiments showed that electrical connectivity is a significant determinant of propagation severity **[Ref 14]**.

The strongest independent evidence for the value of true open-circuit rest comes from a ten-year experimental study published in the Journal of Power Sources (2025): one hundred commercial $LiFePO_4$/C cells stored in an uninterrupted open-circuit state — electrically disconnected, at 50% SOC and 6°C — retained 96–98% of their initial capacity after ten years, with cell-to-cell variation remaining within the range of fresh production batches **[Ref 16]**. Open-circuit rest at a managed SOC is, on the published evidence, the gentlest condition in which a lithium-ion cell can exist. Conventional battery systems cannot give their cells that condition, because the series string remains internally connected and loaded by system electronics at rest.

The Sthoram BMS Cell Isolation function gives every cell exactly that condition, automatically, after every use. Once Auto-Balancing (Function 1, Volume I) drives the

inter-cell differential to zero measured difference, the Sthoram BMS disconnects every cell from every other cell across the entire 24S string. Each cell rests in complete electrical isolation — no external current path, no shared drain, no stacked voltage, no electrical fault-propagation pathway — while per-cell voltage monitoring continues independently throughout. This is not a software parameter or control threshold; it is a structural property of the Sthoram BMS switching architecture, protected by 25 patent applications accepted in India on 13 June 2026, and it is verifiable on the physical 24S Proof of Concept system by any qualified engineer with standard instruments.

What this means in practice: Calendar aging cannot be switched off — chemistry does not stop. What can be switched off is everything the pack adds on top of it: the slow uneven drain of connected electronics, the stacked high voltage, and the electrical pathways a fault can use. Isolation removes all of it, and lets each cell rest the way the ten-year study proved cells rest best: disconnected, at the right state of charge.

1. Introduction

Battery management literature has focused overwhelmingly on the active states — charge and discharge. The idle state has received comparatively less attention, largely because it appears, superficially, to be a period of inactivity: no current is demanded, no charger is connected, the battery appears to be at rest.

That appearance is misleading, and it is worth being exact about why. Inside every cell, connected or not, parasitic electrochemical reactions at the electrode–electrolyte interface consume active lithium and grow the SEI — calendar aging — at a rate set by temperature and SOC **[Ref 01, Ref 02]**. Around the cells, in a conventional pack, the system itself adds three further processes that operate only because the string is connected: monitoring and balancing electronics draw small quiescent currents from the cells — typically from cell groups rather than uniformly — so cells drain at different rates and drift apart in SOC over long rests **[Ref 03, Ref 12]**; the full series voltage remains stacked across the internal string at every moment, sustaining insulation stress and fault energy **[Ref 13]**; and the electrical continuity of the string means the pack's stored energy remains organised as one high-voltage source rather than as twenty-four small, independent ones **[Ref 14]**.

None of these system-level processes is necessary. They persist in conventional architectures because conventional architectures have no switching fabric capable of disconnecting individual cells. The Sthoram BMS Cell Isolation function removes them with a single architectural decision: after balancing completes, every cell is disconnected from every other cell. No inter-cell current path. No shared drain. No stacked voltage. No electrical fault-propagation pathway. Each cell rests in true open-circuit isolation — the storage condition the ten-year shelf-life literature identifies as optimal **[Ref 16]** — monitored individually until the next operating cycle begins.

What this means in practice: In a conventional battery system, the battery never truly rests — the electronics keep sipping from it, the high voltage stays present, and every cell remains wired to every other. Sthoram BMS gives each cell genuine rest: isolated, monitored, and protected.

2. The Connected Idle State: The Hazards, Stated Precisely

This section separates what happens inside every cell regardless of architecture (Sections 2.1–2.2) from what happens only because the string is connected (Sections 2.3–2.5). The distinction matters: isolation is a remedy for the second category, and this paper claims nothing more.

2.1 Calendar Aging — Degradation That Never Stops

Calendar aging is the degradation of a lithium-ion battery as a function of time, independent of cycling. It occurs under open-circuit conditions at rest and is driven primarily by parasitic side reactions at the electrode–electrolyte interface — chiefly the continuous growth of the SEI on the graphite anode. These reactions proceed continuously in any electrolyte-wetted cell; they cannot be stopped by any electrical architecture, isolated or connected. They can only be slowed — by managing the conditions that set their rate.

The severity is quantified in the literature. A Pseudo-Two-Dimensional (P2D) electrochemical simulation study, validated against experimental data across five temperature–SOC conditions, modelled calendar aging in $LiFePO_4$/graphite cells and found SEI thickness exceeding 300 nm and electrolyte conductivity loss exceeding 20% after 36 months at the most severe condition simulated, 55°C and 90% SOC **[Ref 06 — simulation study, experimentally validated]**. The dominant mechanism is loss of lithium inventory (LLI): parasitic reactions at the anode surface consume Li^+ ions irreversibly, locking them into SEI compounds. Experimental studies of LFP/graphite calendar aging consistently identify SEI growth as the dominant contributor **[Ref 01, Ref 07, Ref 08, Ref 09]**.

MIT RESEARCH CORROBORATION The theoretical foundation for exactly this behaviour was established at MIT. Pinson & Bazant (Journal of The Electrochemical Society, 2013) showed that a simple single-particle model of solvent-diffusion-limited SEI growth accurately explains observed capacity fade in commercial graphite-anode cells and predicts future fade from limited accelerated-aging data, with SEI thickness growing continuously with the square root of time — including during storage **[Ref 17]**. This is the canonical mathematical description of why capacity fade accumulates at rest: the reaction never stops; it only slows as the layer thickens.

KEY FINDING [Ref 01, 07] Calendar aging is not a cycling phenomenon. SEI growth proceeds continuously at the electrode–electrolyte interface even under open-circuit conditions at rest, and the resulting capacity fade is dominated by irreversible loss of cyclable lithium. Because idle periods exceed operating periods in most applications, this idle-interval degradation has a substantial impact on total battery life. Isolation does not stop this chemistry — nothing does — but it ensures the cell rests under the external conditions in which that chemistry does the least damage (Sections 2.2, 3.5).

What this means in practice: Every hour any battery sits idle, it loses a tiny, irreversible fraction of its capacity — connected or not. What a system can control is how fast: the resting charge level, and whether the pack's own electronics are quietly adding drain and drift on top. Those are the levers Sthoram BMS pulls.

2.2 SOC Dependency: The Resting Charge Level Sets the Aging Rate

Calendar aging rate is not constant. It depends strongly and non-linearly on the state of charge at which cells rest. Higher SOC means a lower (more lithiated) anode potential, a stronger thermodynamic driving force for electrolyte reduction, and therefore faster SEI growth. Experimental work in Applied Sciences (2025) confirmed for LFP cells that calendar aging is strongly governed by SOC and temperature, with higher SOC and elevated temperature accelerating capacity fade via enhanced SEI growth **[Ref 02]**. The foundational JES study further showed the dependence is stepped rather than smooth, with plateau regions tied to the graphite staging behaviour — storing below the critical anode-stage SOC substantially prolongs life **[Ref 01]**. Modelling and experimental studies of LFP/graphite calendar aging report degradation rates at the most severe storage conditions — high temperature combined with high SOC — up to several times (in the cited modelling, on the order of five times) higher than at optimal low-temperature, moderate-SOC conditions **[Ref 06 — simulation; Ref 10]**. A complementary experimental study in the Journal of Materials Chemistry A (2025) confirmed that calendar aging at high SOC accelerates side reactions, promotes SEI thickening, induces interfacial inhomogeneity, and triggers structural disorder that persists and worsens under subsequent cycling **[Ref 11]**.

MIT RESEARCH CORROBORATION Independent kinetic evidence for the potential-dependence of SEI growth comes from a joint MIT–Stanford experimental programme (Attia, Das, Harris, Bazant & Chueh, Journal of The Electrochemical Society, 2019): using an electroanalytical method on carbon-black negative electrodes, the study measured SEI growth as a strong, non-linear function of electrode potential, current magnitude and current direction **[Ref 18]**. The electrode potential is set by the state of charge — which is why the SOC at which a cell rests is a genuine, physically grounded control lever for the rate at which its SEI grows.

KEY FINDING [Ref 02, 06, 10] Cells stored at high temperature and high SOC degrade several times faster than cells stored at moderate SOC and low temperature. Sthoram BMS SOC Reserve Management (Function 5) keeps cells resting inside the favourable SOC window; Cell Isolation then ensures they hold that SOC — no external drain path exists through which the resting charge level can drift.

What this means in practice: A battery stored full in a warm room is aging several times faster than it needs to. Sthoram BMS parks the cells at the right charge level, then disconnects everything that could change it.

2.3 Uneven External Drain and Divergence Across Long Rests

Precision matters here, and this paper states the physics plainly. In a series string whose terminals are open, no loop current flows: series cells at rest do not exchange charge with one another through the series interconnect. (Charge redistribution between cells at rest is a property of parallel connections, where cells share terminals.) The divergence hazard in a conventional pack at rest arises from three real mechanisms instead. First, every cell self-discharges internally at its own rate — driven by parasitic interface reactions and micro-short paths, strongly dependent on temperature and SOC — and those rates differ from cell to cell **[Ref 12 — experimental characterisation]**. Second, the pack's own electronics load the cells: monitoring ICs, sense networks, and balancing circuitry draw small quiescent currents, typically from cell taps or cell groups rather than perfectly uniformly, adding an uneven external drain on top of the uneven internal one. Third, Monte Carlo simulation of a large 168S pack confirmed that differing self-discharge and intrinsic aging rates produce voltage drift that accumulates across successive rest periods in series strings **[Ref 03 — simulation study]**. Survey literature on cell balancing likewise identifies idle-period divergence as a standing problem that balancing must repeatedly re-correct **[Ref 04 — survey preprint]**.

In the isolated state, the second mechanism is removed entirely and the consequences of the first are contained. Each cell's internal self-discharge continues — isolation does not claim otherwise — but no external circuit loads any cell, no shared electronics drain cell groups unevenly, and the balanced state established by Auto-Balancing is disturbed only by the slow intrinsic chemistry of each individual cell, not by the system around it.

What this means in practice: In a conventional pack at rest, the pack's own electronics keep sipping current from the cells — and they don't sip evenly, so the cells drift apart over weeks of storage. Isolation unplugs everything: each cell keeps its own charge, subject only to its own slow internal chemistry.

2.4 Deep Over-Discharge and Permanent Cell Destruction

Peer-reviewed research published in Scientific Reports (Nature Publishing Group, 2016) investigated the entire over-discharge process of large-format lithium-ion cells, noting that series-connected cells are particularly prone to over-discharge because the string current is common to all cells while their capacities are not. The study found that deep over-discharge beyond approximately −12% SOC produces a characteristic voltage plateau at which copper current-collector dissolution begins: Cu ions migrate through the electrolyte and deposit on the electrodes, forming an irreversible internal short circuit — a permanent failure mode with no recovery pathway **[Ref 05]**.

The scope of that finding must be stated exactly: over-discharge is driven by current flow. A cell at true open circuit cannot discharge itself past 0% SOC into the copper-dissolution regime; the hazard in the idle state exists wherever a sustained current path remains

connected across the string — quiescent system loads, auxiliary electronics, insulation leakage, or a monitoring fault that leaves a drain in place. Over a long unattended rest, such a path steadily depletes the string, and the weakest cell — lowest capacity, highest self-discharge — reaches the danger threshold first, while the remaining cells continue to drive current through it toward reversal. This is precisely the scenario of long-term storage with a connected load that Ref 05's mechanism makes catastrophic.

In the isolated state, this pathway is structurally closed rather than procedurally avoided. No external current path exists across any cell. No system load, leakage path, or neighbouring cell can drive the weakest cell downward. Each cell's worst case at rest is its own internal self-discharge — a slow decline toward low SOC, not a driven excursion into reversal.

CRITICAL FINDING [Ref 05] Deep over-discharge beyond approximately −12% SOC triggers copper current-collector dissolution and irreversible internal short-circuit formation (Scientific Reports, 2016). The mechanism requires a driven current path. Cell Isolation removes every external current path at rest, structurally precluding driven over-discharge of the weakest cell during storage.

What this means in practice: A battery left in storage with anything still connected — even just its own electronics — is slowly being drained, and the weakest cell hits the bottom first, where permanent damage begins. Isolation unplugs the drain. A weak cell in an isolated string simply holds a lower charge. It is not destroyed.

2.5 Fault Energy and the Electrical Dimension of Propagation

The definitive institutional study of failure propagation in multi-cell batteries was performed by Sandia National Laboratories (Lamb et al., Journal of Power Sources, 2015). Its findings are reported here exactly, because they discipline what isolation can and cannot claim. In modules of cylindrical cells — limited thermal contact between neighbours — electrical configuration was decisive: the series (10S1P) module did not propagate a single-cell thermal runaway, while the parallel (1S10P) module suffered an energetic runaway that consumed the module, because parallel-connected neighbours dump their stored energy directly into a failing cell. In modules of stacked pouch cells — large-area thermal contact — propagation consumed the entire module within 60–80 seconds in all configurations tested, series and parallel alike: with strong thermal coupling, heat transfer dominates and electrical configuration matters less **[Ref 14]**. Recent review literature likewise treats thermal transfer and electrical energy supply as distinct contributions to propagation **[Ref 15]**.

The honest conclusions for architecture follow directly. First, electrical isolation eliminates the electrical contribution to propagation: an isolated cell's fault can draw on no neighbour's stored energy through any circuit, and the parallel-type energy-dump

mechanism Sandia observed cannot exist in any form. Second, isolation collapses the fault energy accessible at any point: in a connected 24S string the stacked ~77 V and the full string's charge stand behind every insulation surface and every potential external short **[Ref 13]**; in the isolated state the maximum electrical source anywhere is a single ~3.2 V cell. Third — stated without hedging — isolation does not stop heat. Thermal propagation through physical contact, the dominant mechanism in tightly packed formats, must be addressed by thermal means; in the Sthoram architecture that is the role of Nitrogen Fire Suppression (Function 4), which addresses the atmospheric and thermal dimension while isolation addresses the electrical one.

CRITICAL SAFETY FINDING [Ref 14, 15] Sandia's module experiments showed electrical connectivity is a significant determinant of propagation severity — parallel connectivity fed a module-consuming runaway, and tightly packed pouch modules propagated in 60–80 seconds through thermal contact regardless of configuration. Isolation removes the electrical propagation pathway and reduces accessible electrical fault energy from a ~77 V string to a single ~3.2 V cell; the thermal pathway is addressed separately by Function 4. Neither mechanism is asked to do the other's job.

What this means in practice: A fire needs fuel. Isolation takes away the electrical fuel — no failing cell can draw on its neighbours' energy, and there is no 77-volt string to arc or short. Heat spreading cell-to-cell is a different problem, and Sthoram BMS assigns it to a different tool: the nitrogen suppression system.

3. What Complete Cell Isolation Produces

3.1 Zero External Standby Drain

In a conventional pack at rest, the cells carry the pack: monitoring electronics, sense networks, balancing circuitry and auxiliary loads draw continuous quiescent current from the string or from cell groups. Over days and weeks of idle time this external drain removes stored energy that performed no work, and — because it is rarely uniform across cell taps — it removes it unevenly. Intrinsic self-discharge, by contrast, is a property of the cells themselves: experimental characterisation shows it is driven by parasitic interface reactions, strongly dependent on temperature and SOC **[Ref 12]**, and no architecture eliminates it.

Isolation therefore claims exactly this: in the isolated state, external standby drain on every cell is zero. No system load, no sense-network leakage through the switching fabric, no balancing leakage. Each cell retains the energy it held at isolation, subject only to its own intrinsic self-discharge. Every unit of energy the system did not itself consume is present and available at the start of the next discharge event.

What this means in practice: A conventional battery at rest is like a flask with a very slow leak you cannot patch (the cell's own chemistry) plus several small taps left slightly open (the pack electronics). Isolation cannot fix the flask's nature — but it closes every tap.

3.2 Fault Containment at the Individual Cell Level

When all cells are electrically isolated, the maximum boundary of any electrical fault is the individual cell in which it originates. A developing internal short circuit in one cell can draw only on the energy stored within that cell; no circuit exists through which the other 23 cells can feed it. This is a structural change in the electrical fault physics, consistent with the Sandia finding that electrical connectivity governs the electrical contribution to propagation severity **[Ref 14]**. Thermal containment is the province of Nitrogen Fire Suppression (Function 4): isolation removes the electrical energy pathway, nitrogen suppression removes the atmospheric conditions for combustion, and together they address the electrical and chemical dimensions of a thermal event at the individual-cell level.

What this means in practice: In an isolated system, one cell's electrical problem stays one cell's problem — the other 23 cells are not wired to it and cannot feed it. The heat side of the equation is handled by the suppression system, not left to hope.

3.3 Collapse of Pack-Level Electrical Fault Energy

In a fully connected 24S string, the full string voltage — approximately 77 V at the nominal $LiFePO_4$ cell voltage of 3.2 V — is continuously present across the internal series chain, and stands behind any insulation failure, service contact, or external short circuit at pack level. Experimental characterisation of external short-circuit faults in EV battery packs documents the severity of such events at pack scale **[Ref 13]**, and the available fault current for a given fault resistance scales with the source voltage behind it — elementary circuit physics that needs no citation. In the isolated state, no continuous high-voltage series string exists anywhere in the system at rest: the maximum voltage present between any two accessible points is that of a single isolated cell, approximately 3.2 V. The catastrophic energy release associated with a pack-level short circuit in a connected high-voltage string is structurally precluded for the duration of the isolated state.

What this means in practice: A fully isolated battery at rest is not a high-voltage system. It is 24 individual 3.2-volt cells that happen to share an enclosure. You cannot get a 77-volt short circuit from a system in which no 77-volt string exists at that moment.

3.4 Preservation of the Balanced State for the Next Cycle

The Sthoram BMS sequence is defined and intentional: Auto-Balancing (Function 1) first drives the inter-cell differential to zero measured difference; Cell Isolation (Function 3)

then locks that state in by removing every external path through which it could erode. Research in the Journal of Materials Chemistry A (2025) confirmed that degradation accumulated during poorly managed rest persists and worsens under subsequent cycling **[Ref 11]** — the quality of the idle period shapes the quality of every cycle that follows. Isolation ensures the next cycle begins from the equilibrium the balancing cycle verified, altered only by each cell's slow intrinsic self-discharge rather than by weeks of uneven system drain.

What this means in practice: Balancing without isolation is like carefully sorting a deck of cards, then leaving it on a table where the draft can shuffle it. Balancing with isolation seals the sorted deck in a box. When you open the box, the order is as you left it — apart from what time itself does, which no box prevents.

3.5 Long-Term Storage: The Open-Circuit Evidence

The most direct published evidence on optimal rest conditions is the ten-year shelf-life study in the Journal of Power Sources (2025): one hundred commercial 26650 $LiFePO_4$/C cells stored in an uninterrupted open-circuit state at 50% SOC and 6°C retained 96–98% of their initial capacity after ten years, with negligible resistance change, cell-to-cell variation within the range of fresh production batches, unchanged high-rate capability, and aging attributed primarily to SEI growth **[Ref 16]**. It bears emphasis that these cells were not resting inside a connected pack: they were electrically disconnected — precisely the condition Cell Isolation reproduces inside a deployed system. Isolation, combined with SOC Reserve Management (Function 5) holding cells in the favourable SOC window, brings a deployed 24S pack as close to the storage condition of that study as a fielded system can come: open-circuit rest at managed SOC, with the two remaining aging levers — temperature and SOC — actively managed rather than left to circumstance.

What this means in practice: The best long-term storage result ever published for LFP cells — 96–98% capacity after ten years — was achieved by cells resting disconnected at a managed charge level. Conventional packs cannot give their cells that condition. Sthoram BMS builds it in as the default idle state.

4. The State of the Art: Conventional Architectures Do Not Isolate Cells at Rest

A fair description of conventional practice is as follows. Most deployed battery systems do open their main contactors at rest, disconnecting the pack terminals from the external load — pack-level isolation is standard engineering. What conventional architectures do not do is disconnect the cells from each other or from the system's own electronics: behind the contactors, the series string remains continuously interconnected, the monitoring and balancing electronics remain connected across cells or cell groups, and

the full stacked string voltage remains present across the internal chain. There is, in the conventional topologies documented in the literature surveyed for this paper, no switching fabric that places individual cells at open circuit in the idle state. Readers are invited to test this statement against any commercial BMS specification of their choosing.

The documented consequences of that architecture at rest, each within the scope established in Section 2:

- Calendar aging proceeds in all cells at a rate set by SOC and temperature **[Ref 01, 02, 07]** — unavoidable in any architecture — while the connected system adds continuous quiescent drain that can shift cells away from their intended resting SOC over long storage.
- Uneven external drain and differing intrinsic self-discharge rates drive SOC divergence that accumulates across successive rest periods in series strings **[Ref 03 — simulation; Ref 12 — experimental]**, eroding whatever balance the previous balancing cycle achieved **[Ref 04 — survey]**.
- Any sustained current path left across the string during extended storage steadily depletes it, and the weakest cell reaches deep over-discharge first — the regime in which copper dissolution forms a permanent internal short **[Ref 05]**.
- The electrical contribution to fault propagation remains available: Sandia's experiments showed electrical connectivity is a significant determinant of propagation severity, with parallel connectivity feeding a module-consuming runaway, and thermally coupled pouch modules propagating in 60–80 seconds regardless of configuration **[Ref 14]**.
- The full pack voltage — approximately 77 V for a 24S LFP string — is continuously present across the internal series chain, maintaining insulation stress and external-short fault energy at all times **[Ref 13]**.
- No commercial BMS specification reviewed for this paper includes individual-cell electrical isolation during the idle state as a standard operating function.

What this means in practice: Conventional systems disconnect the battery from the world, but never from itself. To the knowledge of the authors, Sthoram BMS is the first architecture that disconnects every cell from every other cell — and from the system's own electronics load — after balancing, as the standard idle state.

5. Cell Isolation: Operating State Specification

5.1 Sequence and Trigger

Cell Isolation is not triggered at the moment of entering idle. It is the terminal phase of the idle-state entry sequence:

Step 1	The battery string transitions to the idle state (no charge source connected, no

	load connected).
Step 2	Auto-Balancing (Function 1) is activated automatically under Master Brain command. The switching fabric reconfigures the cell interconnect path to redistribute energy between cells until the inter-cell voltage differential converges to zero measured difference. Maximum duration: 2 hours; in practice, equilibrium is typically reached earlier.
Step 3	Once zero measured differential is confirmed by continuous per-cell voltage monitoring, the switching fabric disconnects every cell from every other cell across the entire 24S string.
Step 4	Each of the 24 cells rests in complete electrical isolation. No external current path exists across any cell, and the series string voltage is not present across the pack terminals or across the internal chain.
Step 5	Voltage monitoring of each cell continues independently throughout the isolation state via the per-cell monitoring architecture. Each cell's voltage is observable individually in real time through the Master Brain and the optional mobile application.

This sequence ensures that isolation is always entered from a balanced state. Cells are never isolated while in an imbalanced condition: the zero-differential confirmation is the prerequisite, not the consequence, of isolation.

5.2 Monitoring During Isolation

Per-cell voltage monitoring continues throughout the isolation state. The mechanism by which continuous monitoring is achieved without constituting a shared current path between cells is part of the patented switching architecture (25 applications accepted in India, 13 June 2026) and is not disclosed in this document. What is claimed — and is directly verifiable on the physical system — is the observable outcome: each cell's voltage is individually readable in real time throughout the isolation period, and the isolation properties specified in Section 5.3 hold while monitoring is active. Independent verifiers are encouraged to test both claims simultaneously.

5.3 Proof of Concept: Independent Verification Protocol

The Cell Isolation function is independently verifiable on the Sthoram BMS 24S Proof of Concept battery pack by any qualified engineer, research institution, or testing body, using standard instruments. The verifiable observables, stated in measurable terms:

- **Open circuit between cells.** During the isolation state, a resistance or continuity measurement across the interconnect path between any two adjacent cells shows an open circuit — no conductive path. Equivalently, an ammeter inserted in any inter-cell interconnect during isolation reads zero current.

- **No series string at the terminals.** The pack terminal voltage in the isolated state does not present the series string voltage: the ~77 V stack of the connected 24S configuration is absent, confirming that no continuous high-voltage string exists.
- **Charge state preserved.** Each cell holds the voltage at which it was isolated — zero measured differential from every other cell, as established by the preceding Auto-Balancing cycle — with subsequent change attributable only to intrinsic per-cell self-discharge over time.
- **Monitoring alive.** Individual cell voltages remain observable in real time through the Master Brain system and the optional mobile application throughout the isolation period.

VERIFICATION CLAIM — OPEN TO ALL Every property above is measurable with standard instruments on the physical 24S POC system — continuity between cells, pack terminal voltage, per-cell voltages over time. The claims are stated in instrument-readable terms precisely so that they can be checked by any party, at any time, without reliance on the authors.

What this means in practice: You can walk up to this battery while it is idle, measure between any two adjacent cells, and find an open circuit — no current path at all. Measure the pack terminals and the 77-volt string simply is not there. That is what isolation means, and that is what you will see.

6. Summary of Documented Outcomes and Their Sources

The findings below are drawn from the sources cited in Section 8, with the nature of each source stated. Peer-reviewed experiments, simulations, surveys and reviews are different classes of evidence and are labelled accordingly.

- **Calendar aging is a continuous idle-state degradation mechanism.** SEI growth proceeds at the electrode–electrolyte interface under open-circuit conditions, with capacity fade dominated by loss of cyclable lithium; fade depends on SOC in plateau-stepped fashion tied to graphite staging. (Peer-reviewed experiment, JES 2016) **[01]**
- **Resting SOC and temperature set the aging rate.** Higher SOC and temperature accelerate LFP capacity fade via enhanced SEI growth. (Peer-reviewed experiment, 2025) **[02]**
- **Severe idle conditions produce severe, quantified degradation.** SEI thickness exceeding 300 nm and electrolyte conductivity loss over 20% after 36 months at 55°C / 90% SOC. (Experimentally validated P2D simulation, 2025) **[06]**
- **Divergence accumulates across rests in series strings.** Differing self-discharge and intrinsic aging rates produce voltage drift that accumulates across successive rest periods. (Peer-reviewed Monte Carlo simulation of a 168S pack, 2020) **[03]** — with self-discharge mechanisms characterised experimentally in **[12]**.

- **High-SOC rest causes damage that persists into cycling.** Side-reaction acceleration, SEI thickening, interfacial inhomogeneity and structural disorder that worsen under subsequent cycling. (Peer-reviewed experiment, 2025) **[11]**
- **Deep over-discharge permanently destroys cells.** Beyond approximately −12% SOC, copper collector dissolution forms an irreversible internal short circuit; the mechanism requires a driven current path. (Peer-reviewed experiment, Scientific Reports 2016) **[05]**
- **Electrical connectivity shapes propagation severity.** A series cylindrical module did not propagate a single-cell runaway; a parallel module was consumed; thermally coupled pouch modules propagated in 60–80 seconds in all configurations. (Institutional experiment, Sandia / JPS 2015) **[14]** — with thermal and electrical contributions treated as distinct mechanisms in recent review literature **[15]**.
- **Open-circuit rest at managed SOC preserves nearly all capacity.** 96–98% retention after ten years of uninterrupted open-circuit storage at 50% SOC and 6°C, with cell-to-cell variation within fresh-batch range. (Peer-reviewed ten-year experiment, 2025) **[16]**
- **SEI growth at rest has a canonical mathematical theory, developed at MIT.** Solvent-diffusion-limited SEI growth explains capacity fade in commercial graphite-anode cells and enables lifetime prediction; SEI growth strongly depends on electrode potential. (Peer-reviewed theory and experiment, MIT / MIT–Stanford) **[17, 18]**

Aspect (source type)	Connected Idle (Conventional)	Isolated Idle (Sthoram BMS)
Calendar aging — intra-cell chemistry [Ref 01, 02, 06]	Proceeds at a rate set by SOC and temperature; system drain can shift resting SOC over long storage	Proceeds identically as chemistry — but at the managed SOC, which no external drain can shift
External standby drain on cells [system architecture]	Continuous quiescent load from monitoring/balancing electronics, typically uneven across cell groups	Zero — no external circuit loads any cell; only intrinsic self-discharge remains **[Ref 12]**
SOC divergence across rest periods [Ref 03 — simulation; Ref 12]	Uneven drain plus differing intrinsic self-discharge accumulate drift across successive rests	Balanced state preserved; change bounded by each cell's own intrinsic self-discharge
Driven deep over-discharge of weakest cell [Ref 05]	Possible wherever a sustained current path remains across the string in storage — Cu dissolution beyond ~−12% SOC is permanent	Structurally precluded — no external current path exists across any cell
Electrical dimension of fault propagation [Ref 14, 15]	Available: connectivity determines the electrical contribution; parallel paths feed failing cells; ~77 V string stands behind faults **[Ref 13]**	Removed: electrical fault boundary is the single cell (~3.2 V); thermal pathway handled by Function 4

Long-term storage condition vs. published optimum [Ref 16]	Cells rest inside a loaded, interconnected system — a condition no shelf-life study recommends	Open-circuit rest at managed SOC — the condition under which 96–98% ten-year retention was demonstrated

7. Conclusion

The scientific evidence is consistent and independently established: the idle state is not a period of electrochemical rest. Calendar aging proceeds continuously inside every cell, at a rate set by SOC and temperature **[Ref 01, 02, 06]** — that much no architecture can change, and this paper does not claim otherwise. What architecture does control is everything the system adds around that chemistry: the quiescent drain and uneven loading of connected electronics, the divergence that accumulates across rests **[Ref 03, 12]**, the possibility of driven deep over-discharge wherever a current path remains across the string in storage **[Ref 05]**, the electrical dimension of fault propagation **[Ref 14, 15]**, and the continuous presence of the full ~77 V stacked string behind every insulation surface **[Ref 13]**.

Complete electrical isolation of each cell at rest removes that entire system-level layer at once. It preserves the balanced state established by Auto-Balancing, closes every external drain and over-discharge path, collapses accessible electrical fault energy from a 77 V string to a single 3.2 V cell, and gives every cell the storage condition — open-circuit rest at managed SOC — under which the published ten-year record of 96–98% capacity retention was achieved **[Ref 16]**. Thermal propagation, the dimension isolation does not address, is assigned explicitly to Nitrogen Fire Suppression (Function 4).

In the conventional topologies documented in the literature surveyed for this paper, no BMS architecture includes individual-cell electrical isolation as a standard idle-state function; conventional systems isolate the pack from the world, not the cells from each other. The Sthoram BMS Cell Isolation function — entered automatically after Auto-Balancing confirms zero measured differential, and maintained with continuous per-cell monitoring — is a structural departure from those topologies, protected by 25 patent applications accepted in India (13 June 2026; international filing eligible from 25 July 2026 under Section 39, Indian Patents Act, 1970).

Every property claimed is stated in instrument-readable terms and is verifiable on the physical 24S Proof of Concept system, by any qualified engineer, with standard instruments, at any time. These are not assertions offered for belief. They are demonstrations offered for measurement.

What this means in practice: To the knowledge of the authors, Sthoram BMS is the first battery management architecture that disconnects every cell from every other cell after balancing, as the standard idle state — verified on a physical 24-cell system, demonstrable to any engineer with standard instruments, and grounded, within their

stated scope, in the same peer-reviewed studies that establish the problems it addresses. We invite the measurement.

8. Bibliography

All references are to peer-reviewed publications, institutional research, or archival preprints; the nature of each source (experiment, simulation, survey, review) is stated in its annotation and respected in the body text. Reference numbering is preserved from Version 2.0 of this paper; entries **[17]** and **[18]** are added in this edition. The DOI of entry **[13]** has been corrected in this edition.

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